

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 26 /03 /2026**
Time: 3 Hrs **GTM - 05** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. If the magnetic field in a plane electromagnetic wave is given by $\vec{B} = 3\times10^{-8}\sin\left(1.6\times10^{3}x + 48\times10^{10}t\right)\hat{j}\ T$; then what will be expression for electric field?

A) $\vec{E} = \left(9\sin\left(1.6\times10^{3}x + 48\times10^{10}t\right)\hat{k}V/m\right)$

B) $\vec{E} = \left(3\times10^{-8}\sin\left(1.6\times10^{3}x + 48\times10^{10}t\right)\hat{i}V/m\right)$

C) $\vec{E} = \left(3\times10^{-8}\sin\left(1.6\times10^{3}x + 48\times10^{10}t\right)\hat{j}V/m\right)$

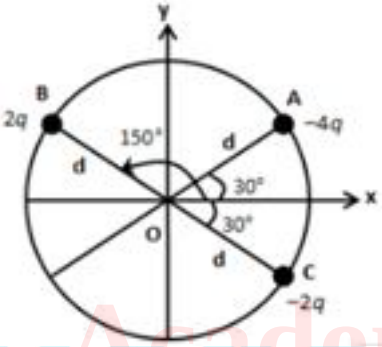
D) $\vec{E} = \left(60\sin\left(1.6\times10^{3}x + 48\times10^{10}t\right)\hat{k}V/m\right)$

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2. A particle of mass m and charge q has an initial velocity $\vec{v} = v_0\hat{j}$. If an electric field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{i}$ act on the particle, its speed will be double after a time.

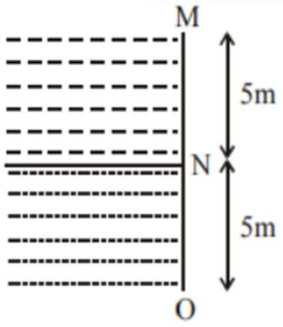
A) $\frac{\sqrt{3}mv_0}{qE_0}$ B) $\frac{2mv_0}{qE_0}$ C) $\frac{3mv_0}{qE_0}$ D) $\frac{\sqrt{2}mv_0}{qE_0}$

3. Three charged particles A, B and C with charges $-4q, 2q$ and $-2q$ are present on the circumference of a circle of radius d. The charged particles A, C and centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x -direction is



A) $\frac{2\sqrt{3}q}{\pi\varepsilon_0 d^2}$ B) $\frac{\sqrt{3}q}{\pi\varepsilon_0 d^2}$ C) $\frac{3\sqrt{3}q}{4\pi\varepsilon_0 d^2}$ D) $\frac{\sqrt{3}q}{4\pi\varepsilon_0 d^2}$

4. Two liquids of densities ρ_1 and $\rho_2(\rho_2 = 2\rho_1)$ are filled up behind a square wall of side 10 m as shown in figure. Each liquid has a height of 5 m. The ratio of the forces due to these liquids exerted on upper part MN to that at the lower part NO is (Assume that the liquids are not mixing)



A) 1/4 B) 1/2 C) 2/3 D) 1/3

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5. Two particles of equal mass m have respective initial velocities $u\hat{i}$ and $u\left(\frac{\hat{i}+\hat{j}}{2}\right)$. They collide completely inelastically. The energy lost in the process is

A) $\frac{3}{4}mu^2$ B) $\frac{1}{8}mu^2$ C) $\frac{1}{3}mu^2$ D) $\sqrt{\frac{2}{3}}mu^2$

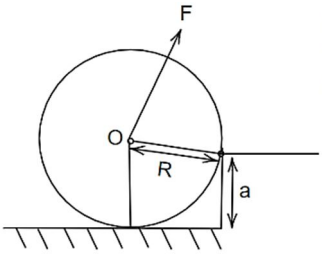
6. A particle starts from the origin at $t=0$ with an initial velocity of $3.0\hat{i}\ m/s$ and moves in the $x-y$ plane with a constant acceleration $\left(6.0\hat{i}+4.0\hat{j}\right)m/s^2$. The X-coordinate of the particle at the instant when its y-coordinate is 32m is D meters. The value of D is

A) 50 B) 32 C) 60 D) 40

7. Train A and train B are running on parallel tracks in the opposite directions with speeds of 36 km/ hour and 72 km/ hour respectively. A person is walking in train A in the direction opposite to its motion with a speed of 1.8 km/hour. Speed (in ms^{-1}) of this person as observed from train B will be close to: (take the distance between the tracks as negligible)

A) 29.5 ms^{-1} B) 28.5 ms^{-1} C) 31.5 ms^{-1} D) 30.5 ms^{-1}

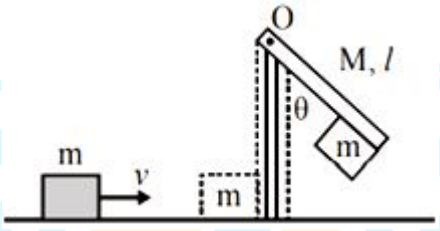
8. A uniform cylinder of mass M and radius R is to be pulled over a step of height $a(a<R)$ by applying a force F at its centre 'O' perpendicular to the plane through the axes of the cylinder on the edge of the step (see figure). The minimum value of F required is



A) $Mg\sqrt{1-\frac{a^2}{R^2}}$ B) $Mg\sqrt{1-\left(\frac{R-a}{R}\right)^2}$ C) $Mg\sqrt{\left(\frac{R}{R-a}\right)^2-1}$ D) $Mg\frac{a}{R}$

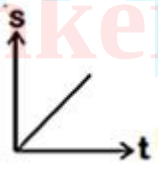
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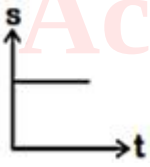
9. A block of mass $m = 1\,kg$ slides with velocity $v = 6\,m/s$ on a frictionless horizontal surface and collides with a uniform vertical rod and sticks to it as shown. The rod is pivoted about O and swings as a result of the collision making angle θ before momentarily coming to rest. If the rod has mass $M = 2\,kg$, and length $l = 1\,m$, the value of θ is approximately : (take $g = 10\,m/s^2$)

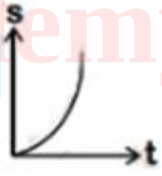


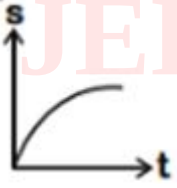
A) 49° B) 55° C) 63° D) 69°

10. A particle is moving unidirectionally on a horizontal plane under the action of a constant power supplying energy source. The displacement (s)- time (t) graph that describes the motion of the particle is (graphs are drawn schematically and are not to scale)

A) 

B) 

C) 

D) 

11. A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50mm, 5.55mm, 5.45mm, and 5.65mm. The average of these four readings is 5.5375 mm and the standard deviation of the data is 0.07395mm. The average diameter of the pencil should therefore be recorded as

A) $(5.5375 \pm 0.0739)\,mm$ B) $(5.5375 \pm 0.0740)\,mm$

C) $(5.54 \pm 0.07)\,mm$ D) $(5.538 \pm 0.074)\,mm$

12. The bulk modulus of a liquid is $3\times10^{10}\,Nm^{-2}$. The pressure required to reduce the volume of liquid by 2% is

A) $3\times10^{8}\,Nm^{-2}$ B) $9\times10^{8}\,Nm^{-2}$ C) $6\times10^{8}\,Nm^{-2}$ D) $12\times10^{8}\,Nm^{-2}$

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13. A stone of mass m, tied to a string is being whirled in a vertical circle with a uniform speed. The tension in the string is

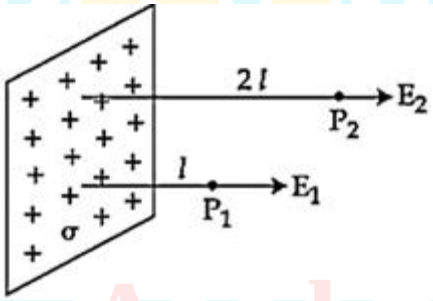
A) The same throughout the motion

B) Minimum at the highest position of the circular path

C) Minimum at the lowest position of the circular path

D) Minimum when the rope is in the horizontal position

14. In the figure, a very large plane sheet of positive charge is shown. P_1 and P_2 are two points at distance ℓ and 2ℓ from the charge distribution. If σ is the surface charge density, then the magnitude of electric fields E_1 and E_2 at P_1 and P_2 respectively are



A) $E_1 = \sigma/\varepsilon_0, E_2 = \sigma/2\varepsilon_0$ B) $E_1 = 2\sigma/\varepsilon_0, E_2 = \sigma/\varepsilon_0$

C) $E_1 = E_2 = \sigma/2\varepsilon_0$ D) $E_1 = E_2 = \sigma/\varepsilon_0$

15. **Assertion A:** Two identical balls A and B thrown with same velocity ' u ' at two different angles with horizontal attained the same range R. If A and B reached the maximum height h_1 and h_2 respectively, then $R = 4\sqrt{h_1 h_2}$

Reason R: Product of said heights. $h_1\, h_2 = \left(\frac{u^2\sin^2\theta}{2g}\right)\cdot\left(\frac{u^2\cos^2\theta}{2g}\right)$

Choose the **correct** answer.

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true but R is NOT the correct explanation of A

C) A is true but R is false.

D) A is false but R is true.

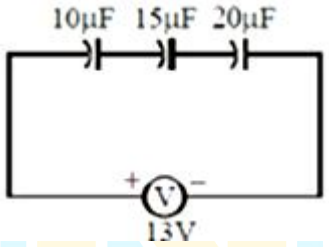
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16. Time period of a simple pendulum in a stationary lift is 'T'. If the lift accelerates with $\frac{g}{6}$ vertically upwards then the time period will be

(Where g = acceleration due to gravity)

A) $\sqrt{\frac{6}{5}}T$ B) $\sqrt{\frac{5}{6}}T$ C) $\sqrt{\frac{6}{7}}T$ D) $\sqrt{\frac{7}{6}}T$

17. The charge on capacitor of capacitance $15\mu F$ in the figure given below is



A) $60\mu C$ B) $130\mu C$ C) $260\mu C$ D) $585\mu C$

18. A longitudinal wave is represented by $x = 10\sin 2\pi\left(nt - \frac{x}{\lambda}\right)cm$. The maximum particle velocity will be four times the wave velocity if the determined value of wavelength is equal to

A) 2π B) 5π C) π D) $\frac{5\pi}{2}$

19. According to kinetic theory of gases,

A. The motion of the gas molecules freezes at $0^{\circ}C$

B. The mean free path of gas molecules decreases if the density of molecules is increased.

C. The mean free path of gas molecules increases if temperature is increased keeping pressure constant.

D. Average kinetic energy per molecule per degree of freedom is $\frac{3}{2}k_BT$ (for monoatomic gases)

Choose the most appropriate answer from the options given below.

A) A and C only B) B and C only C) A and B only D) C and D only

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20. Match List-I with List-II.

List-I		List-II	
A)	Moment of inertia of solid sphere of radius (R) about any tangent.	I)	$\frac{5}{3}MR^2$
B)	Moment of inertia of hollow sphere of radius (R) about any tangent	II)	$\frac{7}{5}MR^2$
C)	Moment of inertia of circular ring of radius (R) about its diameter.	III)	$\frac{1}{4}MR^2$
D)	Moment of inertia of a circular disc of radius (R) about any diameter.	IV)	$\frac{1}{2}MR^2$

Choose the correct answer from the options given below:

A) A-II, B-I, C-IV, D-III

B) A-I, B-II, C-IV, D-III

C) A-II, B-I, C-III, D-IV

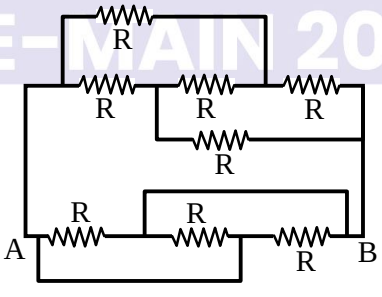
D) A-I, B-II, C-III, D-IV

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. In Young's double slit experiment, 12 fringes are observed by light of λ = 600 nm in a certain segment of the screen. If wavelength is changed to 400 nm then number of fringes in the same segment will be

22. Find equivalent resistance (in ohm) between point A and B in the adjacent circuit. (Each resistor has resistance R = 16 ohm)

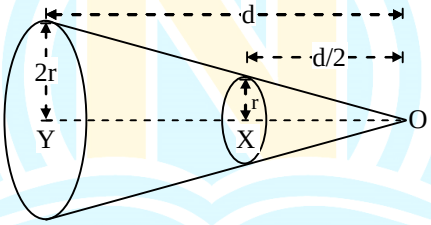


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23. A particle of mass $200\text{MeV}/c^2$ collides with a hydrogen atom at rest. Soon after the collision the particle comes to rest, and the atom recoils and goes to its first excited state. The initial kinetic energy of the particle (in eV) is $\frac{N}{4}$. The value of N is

(Given the mass of the hydrogen atom to be $1\text{GeV}/c^2$) _____.

24. Two circular coils X and Y having equal number of turns and carry equal currents in the same sense and subtend same solid angle at point O. If the smaller coil X is midway between O and Y, then if we represent the magnetic induction due to bigger coil Y at O as B_Y and due to smaller coil X at O as B_X then, $\frac{B_X}{B_Y} =$ ___



25. In a series LR circuit, power of 400 W is dissipated from a source of $250\text{ V}, 50\text{ Hz}$. The power factor of the circuit is 0.8. In order to bring the power factor to unity, a capacitor of value C is added in series to the L and R. Taking the value of C as $\left(\frac{n}{3\pi}\right)\mu\text{F}$, then value of n is _____.

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

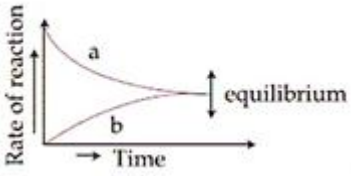
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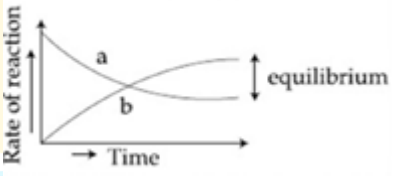
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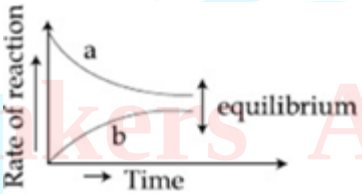
26. The radius of the second Bohr orbit, in terms of the Bohr radius, a_0, in Li^{2+} is

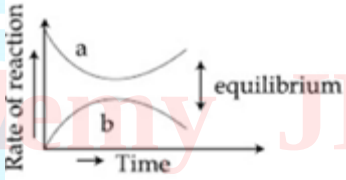
A) $\frac{2a_0}{3}$ B) $\frac{2a_0}{9}$ C) $\frac{4a_0}{9}$ D) $\frac{4a_0}{3}$

27. For the equilibrium $A \rightleftharpoons B$, the variation of the rate of the forward (a) and reverse (b) reaction with time is given by

A) 

B) 

C) 

D) 

28. Given below are the oxides: Na_2O, As_2O_3, N_2O, NO and Cl_2O_7

Number of amphoteric oxides is

A) 0 B) 1 C) 2 D) 3

29. The first ionization enthalpies of Be, B, N and O follow the order

A) $O < N < B < Be$ B) $Be < B < N < O$

C) $B < Be < N < O$ D) $B < Be < O < N$

30. Given two statements below.

Statement- I: In Cl_2 molecule the covalent radius is double of the atomic radius of chlorine.

Statement- II: Radius of anionic species is always greater than their parent atomic radius.

Choose the most appropriate answer from the options given below.

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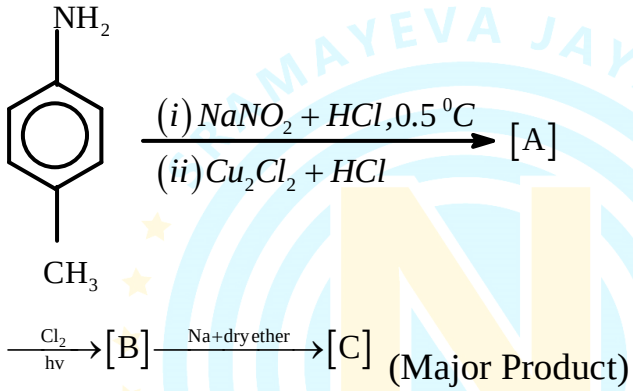
A) Both statement- I and statement-II are correct.

B) Both statement -I and statement- II are incorrect.

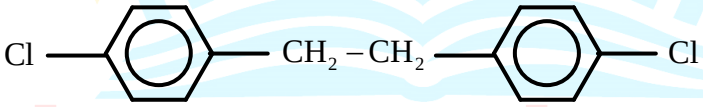
C) Statement -I is correct but statement- II is incorrect.

D) Statement- I is incorrect but statement- II is correct.

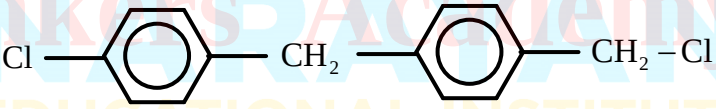
31. In the following reaction sequence, [C] is



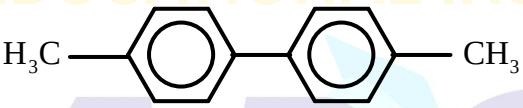
A)



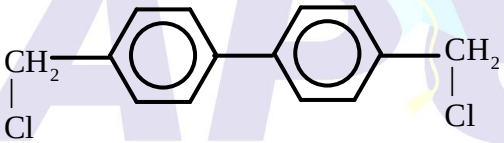
B)



C)

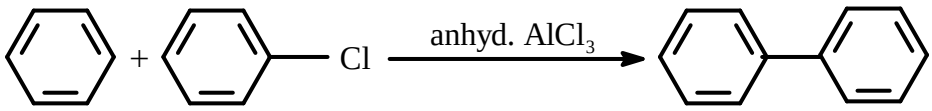


D)

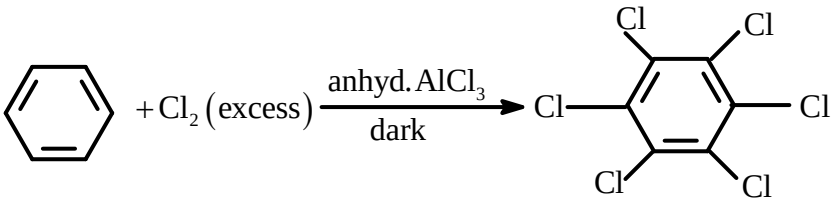


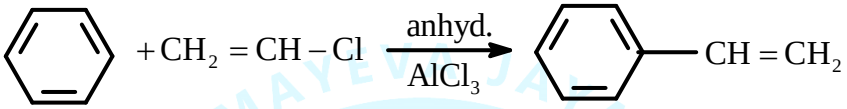
32. Consider the following reactions.

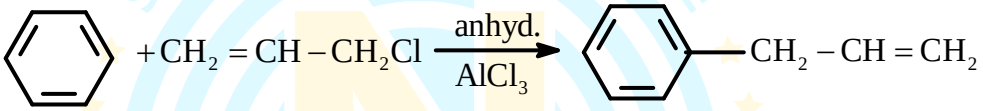
a)



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b) 

c) 

d) 

Which of these reactions are possible?

A) (a) and (d) B) (a) and (b) C) (b), (c) and (d) D) (b)and (d)

33. Which of the following statements are **correct**?

(A) The electronic configuration of Cr is $[Ar]3d^5 4s^1$.

(B) The magnetic quantum number may have a negative value only

(C) In the ground state of an atom, the orbitals are filled in order of their increasing energies.

(D) The total number of nodes are given by $n-2$.

Choose the most appropriate answer from the options given below.

A) (A),(C) and (D) only B) (A) and (B) only

C) (A) and (C) only D) (A), (B) and (C) only

34. The **incorrect** statement is

A) Manganate and permanganate ions are paramagnetic.

B) Manganate ion is green in colour and permanganate ion is purple in colour.

C) In manganate and permanganate ions, the π-bonding takes place by overlap of p-orbitals of oxygen and d-orbitals of manganese.

D) Manganate and permanganate ions are tetrahedral.

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35. Consider the following reactions.

(i) Glucose $+ROH \xrightarrow{\text{dry HCl}}$ Acetal $\xrightarrow{\text{x eq.of } (CH_3CO)_2O}$ acetyl derivative

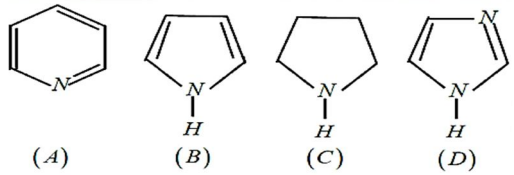
(ii) Glucose $\xrightarrow{Ni/H_2} A \xrightarrow{\text{y eq.of } (CH_3CO)_2O}$ acetyl derivative

(iii) Glucose $\xrightarrow{\text{z eq.of } (CH_3CO)_2O}$ acetyl derivative

'x', 'y' and 'z' in these reactions are respectively.

A) 4,6 & 5 B) 5,6 & 5 C) 5,4 & 5 D) 4,5 & 5

36. The increasing order of basicity of the following compounds is



A) (A) < (B) < (C) < (D) B) (B) < (A) < (D) < (C)

C) (B) < (A) < (C) < (D) D) (D) < (A) < (B) < (C)

37. Lattice enthalpy and enthalpy of solution of NaCl are 788 kJ mol^{-1} and 4 kJ mol^{-1}, respectively. The hydration enthalpy of NaCl is

A) 784 kJ mol^{-1} B) 780 kJ mol^{-1} C) −784 kJ mol^{-1} D) −780 kJ mol^{-1}

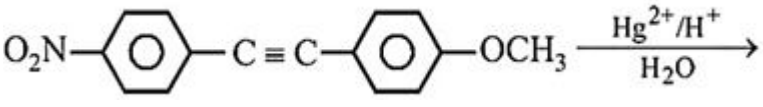
38. Of the species, NO, NO^+, NO^{2+} and NO^-, the one with maximum bond strength is

A) NO B) NO^{2+} C) NO^- D) NO^+

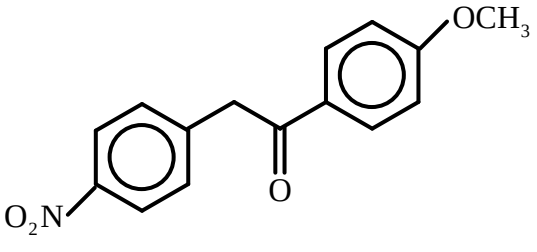
39. The depression in freezing point observed for a formic acid solution of concentration 0.5 mLL^{-1} is $0.0405^\circ C$. Density of formic acid is 1.05 g mL^{-1}. The Van't Hoff factor of the formic acid solution is nearly : (Given for water $k_f = 1.86 kkgmol^{-1}$)

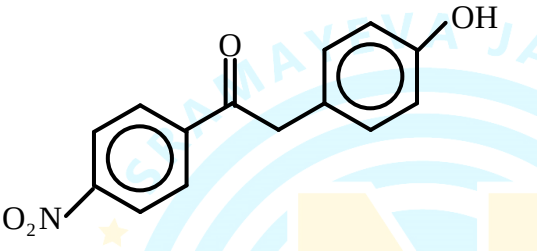
A) 0.8 B) 1.1 C) 1.9 D) 2.4

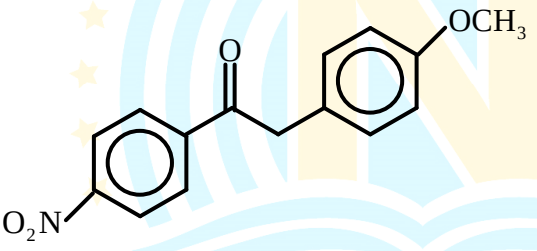
40. The major product obtained from the following reaction is

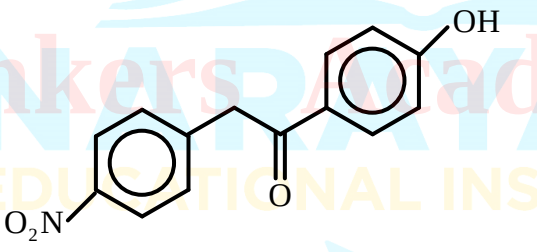


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A) 

B) 

C) 

D) 

41. At $25^{\circ}C$ and 1 atm pressure, the enthalpy of combustion of benzene (l) and acetylene (g) are -3268 kJ mol^{-1} and -1300 kJ mol^{-1}, respectively. The change in enthalpy for the reaction $3C_2H_2(g) \rightarrow C_6H_6(l)$, is

A) +324 kJ mol^{-1} B) +632 kJ mol^{-1} C) -632 kJ mol^{-1} D) -732 kJ mol^{-1}

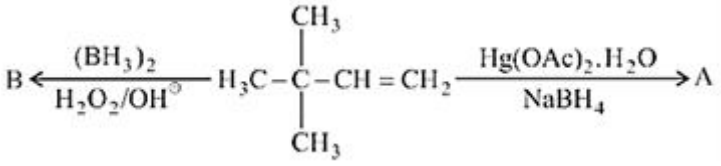
42. The most common oxidation state of Lanthanoid elements is +3. Which of the following is likely to deviate easily from +3 oxidation state?

A) Ce (At. No. 58) B) La (At. No. 57)

C) Lu (At. No. 71) D) Gd (At. No. 64)

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43. Choose the **correct** option for the following reactions.



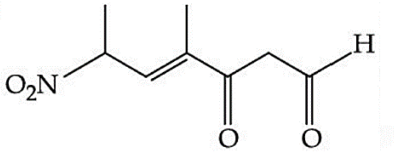
A) 'A' and 'B' are both Markovnikov addition products.

B) 'A' is Markovnikov product and 'B' is anti-Markovnikov product.

C) 'A ' and 'B' are both anti-Markovnikov products.

D) 'B' is Markovnikov and 'A' is anti-Markovnikov products.

44. The **correct** IUPAC name of the following compound is



A) 4-methyl-2-nitro-5-oxohept-3-enal

B) 4-methyl-5-oxo-2-nitrohept-3-enal

C) 4-methyl-6-nitro-3-oxohept-4-enal

D) 6-formyl-4-methyl-2-nitrohex-3-enal

45. A white precipitate was formed when $BaCl_2$ was added to water extract of an inorganic salt. Further, a gas ' X ' with characteristic odour was released when the formed white precipitate was dissolved in dilute HCl. The anion present in the inorganic salt is

A) I^- B) SO_3^{2-} C) S^{2-} D) NO_2^-

SECTION – II
(Numerical Value Answer Type)

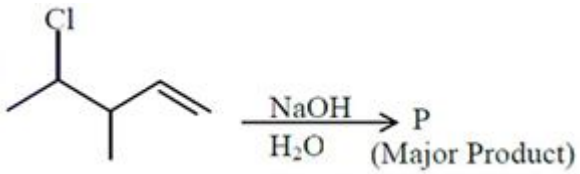
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. Number of grams of bromine that will completely react with 5.0 g of pent-1-ene is _____ $\times 10^{-2}$ g. (Atomic mass of Br = 80 g / mol) [Nearest Integer)

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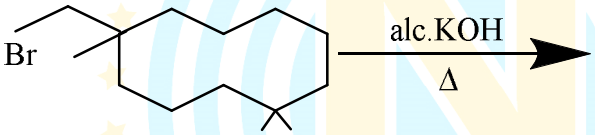
47. The work function of sodium metal is $4.41\times10^{-19}\,J$. If photons of wavelength 300 nm are incident on the metal, the kinetic energy of the ejected electrons will be $\left(h=6.63\times10^{-34}\,Js;c=3\times10^{8}\ m/s\right)\times10^{-21}\ J$.

48.



Consider the above reaction. The number of π electrons present in the product ' P ' is _____.

49.



Product mixture.

Number of possible alkenes formed in product mixture.

50. The atomic number of Unnilunium is _____.

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If $f(a+b+1-x)=f(x)$, for all x, where a and b are fixed positive real numbers, then $\frac{1}{a+b}\int_{a}^{b}x\left(f(x)+f(x+1)\right)dx$ is equal to

A) $\int_{a+1}^{b+1}f(x)dx$ B) $\int_{a+1}^{b+1}f(x+1)dx$ C) $\int_{a-1}^{b-1}f(x+1)dx$ D) $\int_{a-1}^{b-1}f(x)dx$

52. The inverse function of $f(x)=\frac{8^{2x}-8^{-2x}}{8^{2x}+8^{-2x}}, x\in(-1,1)$, is

A) $\frac{1}{4}\log_e\left(\frac{1+x}{1-x}\right)$ B) $\frac{1}{4}\log_e\left(\frac{1-x}{1+x}\right)$

C) $\frac{1}{4}(\log_8 e)\log_e\left(\frac{1-x}{1+x}\right)$ D) $\frac{1}{4}(\log_8 e)\log_e\left(\frac{1+x}{1-x}\right)$

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53. Let z be a complex number such that $\left|\frac{z-i}{z+2i}\right|=1$ and $|z|=\frac{5}{2}$. Then the value of $|z+3i|$ is

A) $\sqrt{10}$ B) $2\sqrt{3}$ C) $\frac{15}{4}$ D) $\frac{7}{2}$

54. Area (in sq. units) of the region outside $\frac{|x|}{2}+\frac{|y|}{3}=1$ and inside the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1$ is

A) $3(4-\pi)$ B) $6(4-\pi)$ C) $6(\pi-2)$ D) $3(\pi-2)$

55. A survey shows that 63% of the people in a city read newspaper A whereas 76% read newspaper B. If x% of the people read both the newspapers, then a possible value of x can be

A) 29 B) 55 C) 65 D) 37

56. The set of all real values of λ for which the quadratic equations, $(\lambda^2+1)x^2-4\lambda x+2=0$ always have exactly one root in the interval $(0,1)$ is

A) $(2,4]$ B) $(-3,-1)$ C) $(1,3]$ D) $(0,2)$

57. The value of $\sum_{r=0}^{20} {}^{50-r}C_6$ is equal to

A) ${}^{50}C_6-{}^{30}C_6$ B) ${}^{50}C_7-{}^{30}C_7$ C) ${}^{51}C_7-{}^{30}C_7$ D) ${}^{51}C_7+{}^{30}C_7$

58. The derivative of $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ with respect to $\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$ at $x=\frac{1}{2}$ is

A) $\frac{2\sqrt{3}}{3}$ B) $\frac{\sqrt{3}}{10}$ C) $\frac{2\sqrt{3}}{5}$ D) $\frac{\sqrt{3}}{12}$

59. If $\int\left(e^{2x}+2e^{x}-e^{-x}-1\right)e^{\left(e^{x}+e^{-x}\right)}dx=g(x)e^{\left(e^{x}+e^{-x}\right)}+c$, where c is a constant of integration, then $g(0)$ is equal to

A) 2 B) 1 C) e D) e^2

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60. If $4\sin 27^{\circ} = \sqrt{\alpha} - \sqrt{\beta}$, then the value of $\alpha + \beta$ is

A) 5 B) 3 C) 8 D) 2

61. The length of the chord $y = \sqrt{3}x - 2\sqrt{3}$ intercepted by the parabola $y^2 = 4(x-1)$ is equal to

A) $4\sqrt{3}$ units B) $\frac{8}{3}$ units C) $\frac{16}{3}$ units D) $\frac{4}{\sqrt{3}}$ units

62. If 2 data sets having 10 and 20 observations have coefficients of variation 50 and 60 respectively and arithmetic means 30 and 25 respectively, then the combined variance of those 30 observation is

A) $\frac{2075}{3}$ B) $\frac{2075}{9}$ C) $\frac{1000}{9}$ D) $\frac{1075}{3}$

63. If $\frac{dy}{dx} + \frac{2^{x-y}\left(2^{y}-1\right)}{2^{x}-1} = 0, x, y > 0, y(1) = 1$, then $y(2)$ is equal to

A) $2 + \log_2 3$ B) $2 + \log_3 2$ C) $2 - \log_3 2$ D) $2 - \log_2 3$

64. Let A and B be any two 3×3 symmetric and skew symmetric matrices respectively. Then which of the following is NOT true?

A) $A^4 - B^4$ is a symmetric matrix

B) $AB - BA$ is a symmetric matrix

C) $B^5 - A^5$ is a skew - symmetric matrix

D) $AB + BA$ is a skew - symmetric matrix

65. If $\lim_{x\to 0} \frac{\alpha e^{x} + \beta e^{-x} + \gamma \sin x}{x\sin^2 x} = \frac{2}{3}$, where $\alpha, \beta, \gamma \in R$, then which of the following is NOT correct?

A) $\alpha^2 + \beta^2 + \gamma^2 = 6$ B) $\alpha\beta + \beta\gamma + \gamma\alpha + 1 = 0$

C) $\alpha\beta^2 + \beta\gamma^2 + \gamma\alpha^2 + 3 = 0$ D) $\alpha^2 - \beta^2 + \gamma^2 = 4$

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66. Let the lengths of the altitudes from the vertices $A(-1,1), B(5,2), C(3,-1)$ of ΔABC are p_1, p_2, p_3 units respectively, then the value of $\dfrac{\left(\frac{1}{p_1}\right)^2+\left(\frac{1}{p_3}\right)^2}{\left(\frac{1}{p_2}\right)^2}$ is equal to

A) 2.5 B) 3.5 C) 4.5 D) 5.5

67. Let the maximum area of the triangle that can be inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{4}=1, a>2$, having one of its vertices at one end of the major axis of the ellipse and one of its sides parallel to the y-axis, be, $6\sqrt{3}$. Then the eccentricity of the ellipse is

A) $\frac{\sqrt{3}}{2}$ B) $\frac{1}{2}$ C) $\frac{1}{\sqrt{2}}$ D) $\frac{\sqrt{3}}{4}$

68. Let S be the set of all $a \in R$ for which the angle between the vectors $\vec{u}=a(\log_e b)\hat{i}-6\hat{j}+3\hat{k}$ and $\vec{v}=(\log_e b)\hat{i}+2\hat{j}+2a(\log_e b)\hat{k}, (b>1)$ is acute. Then S is equal to

A) $\left(-\infty,-\frac{4}{3}\right)$ B) ϕ C) $\left(-\frac{4}{3},0\right)$ D) $\left(\frac{12}{7},\infty\right)$

69. If 3 numbers are selected from the first 15 natural numbers, then the probability that the numbers are in A.P. is

A) $\frac{7}{65}$ B) $\frac{9}{65}$ C) $\frac{8}{65}$ D) $\frac{6}{65}$

70. A plane P =0, which is perpendicular to line $\frac{x-2}{2}=\frac{y+2}{2}=\frac{z-1}{1}$ is passing through the point at which the above line meets the plane $x+y+z=21$, then the distance of plane P= 0 from origin is

A) $\frac{7}{3}$ B) 5 C) $\frac{32}{3}$ D) $\frac{37}{3}$

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
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71. If the line segment joining P (2,3) and Q (5,7) subtends a right angle at R (x, y) and the area of $\Delta PQR = 2$ sq.units, then the maximum number of such point R in xy-plane are _____.

72. The number of 4 letter words (with or without meaning) that can be formed from the eleven letters of the word "EXAMINATION" is _____.

73. The number of terms common to the two A.P.'s $3,7,11,\ldots\ldots,407$ and $2,9,16,\ldots..,709$ is _____.

74. Let the position vectors of points 'A' and 'B' be $\hat{i}+\hat{j}+\hat{k}$ and $2\hat{i}+\hat{j}+3\hat{k}$, respectively. A point ' P ' divides the line segment AB internally in the ratio $\lambda:1(\lambda>0)$. If O is the origin and $\overrightarrow{OB}\cdot\overrightarrow{OP}-3|\overrightarrow{OA}\times\overrightarrow{OP}|^2=6$, then λ is equal to_____

75. Consider the system of equations $x+y+z=1, 2x+3y+2z=1, 2x+3y+\left(a^2-1\right)z=a+1$ then system is inconsistence for $a^2=$_____

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